Problem Statement

Let $n \in \mathbb{N}$. Evaluate the limit from the image "dailyintegral-limit-limit-1x1-2026-07-24.jpg" and find the value of the constant k:

$$\lim_{n\to\infty} n\sin(2\pi e n!) = k\pi$$

Detailed Solution

To evaluate the limit involving the factorial and the mathematical constant e, we begin by utilizing the infinite Maclaurin series expansion for e:

$$e = \sum_{m=0}^{\infty} \frac{1}{m!} = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{n!} + \frac{1}{(n+1)!} + \frac{1}{(n+2)!} + \dots$$

Step 1: Decompose $en!$ into Integer and Fractional Parts

Multiplying the entire series expansion of e by $n!$, we can split the series into a sum of integer terms and a sum of fractional remainder terms:

$$en! = \sum_{m=0}^{n} \frac{n!}{m!} + \sum_{m=n+1}^{\infty} \frac{n!}{m!}$$

Let us define these two parts as M_n and R_n:

$$M_n = \sum_{m=0}^{n} \frac{n!}{m!} = n! + n! + \frac{n!}{2!} + \cdots + 1$$

$$R_n = \sum_{m=n+1}^{\infty} \frac{n!}{m!} = \frac{1}{n+1} + \frac{1}{(n+1)(n+2)} + \frac{1}{(n+1)(n+2)(n+3)} + \dots$$

Because $m \leq n$ for all terms in M_n, every term $\frac{n!}{m!}$ is an integer. Therefore, $M_n \in \mathbb{Z}$ is an integer.

We can now express $en!$ simply as:

$$en! = M_n + R_n$$

Step 2: Simplify the Trigonometric Argument

We substitute $en! = M_n + R_n$ into our sine function:

$$\sin(2\pi en!) = \sin(2\pi(M_n + R_n)) = \sin(2\pi M_n + 2\pi R_n)$$

Since M_n is an integer, $2\pi M_n$ is an integer multiple of 2π. By the 2π-periodicity of the sine function ($\sin(2\pi m + \theta) = \sin(\theta)$ for any $m \in \mathbb{Z}$), the integer portion vanishes completely:

$$\sin(2\pi e n!) = \sin(2\pi R_n)$$

Thus, the target limit simplifies to:

$$\lim_{n\to\infty} n \sin(2\pi e n!) = \lim_{n\to\infty} n \sin(2\pi R_n)$$

Step 3: Analyze the Asymptotics of R_n

Next, we determine the behavior of R_n as $n \to \infty$. Let us factor out $\frac{1}{n+1}$ from the remainder series:

$$R_n = \frac{1}{n+1}\left(1 + \frac{1}{n+2} + \frac{1}{(n+2)(n+3)} + \dots\right)$$

As $n \to \infty$, it is clear that $R_n \to 0$. To find the exact contribution to the limit, we examine the product nR_n:

$$nR_n = \frac{n}{n+1} + \frac{n}{(n+1)(n+2)} + \frac{n}{(n+1)(n+2)(n+3)} + \dots$$

Taking the limit of each term as $n \to \infty$:

$$\lim_{n\to\infty} \frac{n}{n+1} = 1$$

$$\lim_{n\to\infty} \frac{n}{(n+1)(n+2)} = 0$$

All subsequent higher-order terms in the series also converge rapidly to 0. Therefore:

$$\lim_{n\to\infty} nR_n = 1$$

Step 4: Evaluate the Target Limit

We can now rewrite the limit by introducing $2\pi R_n$ to take advantage of the standard trigonometric limit $\lim_{\theta\to 0} \frac{\sin\theta}{\theta} = 1$:

$$\lim_{n\to\infty} n \sin(2\pi R_n) = \lim_{n\to\infty} \left[2\pi \cdot (nR_n) \cdot \frac{\sin(2\pi R_n)}{2\pi R_n}\right]$$

Because $R_n \to 0$ as $n \to \infty$, we have $\lim_{n\to\infty} \frac{\sin(2\pi R_n)}{2\pi R_n} = 1$. Using our product rule for limits:

$$\lim_{n\to\infty} n \sin(2\pi e n!) = 2\pi \cdot \left(\lim_{n\to\infty} nR_n\right) \cdot \left(\lim_{n\to\infty} \frac{\sin(2\pi R_n)}{2\pi R_n}\right)$$

$$= 2\pi \cdot 1 \cdot 1 = 2\pi$$

Step 5: Solve for k

We are given in the problem statement that the limit equals $k\pi$. Equating our derived result with the given expression:

$$2\pi = k\pi$$

Dividing both sides by π yields:

$$k = 2$$

Final Answer:

$$\mathbf{k = 2}$$